Networked Extended Reality since 2023: Closing the Loop between Network and Experience

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Since 2023, networked extended reality (XR) has moved to standalone headsets that stream rendered frames, avatars, point clouds and Gaussian splats over Wi-Fi and 5G-Advanced. This review covers what recent work has built, measured and learned, with a focus on networking and quality of experience (QoE). User studies agree that loss, jitter and stalls hurt more than tens of milliseconds of added delay, but nearly all of them apply static impairments to one user or one pair. Measurements of commercial headsets and platforms show that the application stack often sets what users see, and that standard XR traffic models underestimate real demand. In the radio access, 5G-Advanced makes scheduling aware of video frames through PDU sets, and network-assisted rate control, Wi-Fi 7 multi-link operation and robust mmWave beams follow the same idea. In transport, L4S, Media over QUIC and selective forward error correction aim at frame deadlines rather than throughput, while foveated, split and Gaussian-splat delivery adapt what is sent. We group these approaches by the idea they share, state what each assumes, and show that capacity gains are almost always counted in packet-delay budgets, not in validated QoE. We close with open research questions on replaying real network dynamics to users, on who should own the rate control loop, and on group experience.

CCS Concepts: • **Networks** → **Mobile networks**; **Network performance evaluation**; • **Human-centered computing** → *Virtual reality*.

Additional Key Words and Phrases: extended reality, quality of experience, split rendering, 5G-Advanced, PDU set, Wi-Fi 7, millimetre wave, L4S, Media over QUIC, volumetric video, Gaussian splatting, cloud VR gaming, social XR

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1 INTRODUCTION

Networked extended reality (XR) changed shape after 2023. The standalone headset became the main client: it tracks the user, renders what it can, and streams the rest from an edge or cloud renderer. Split rendering moved from cables to Wi-Fi 6 and 7 and to 5G-Advanced, where 3GPP Releases 18 and 19 added the first radio features designed around XR traffic [41]. The content changed too. Besides rendered frames and 360-degree video, systems now stream point clouds, avatars sent as keypoints [10], and 3D Gaussian splats [26]. The metaverse push added social sessions in which several people share one scene and judge each other's delay.

These changes couple networking and quality of experience (QoE) more tightly than before. A frame is useful only if it arrives before its display deadline, so the network decides whether an image appears, not only how sharp it is. The headset hides head-rotation delay by reprojection, but it cannot hide a lost frame, a frozen peer or a stalled scene. The coupling now also runs the other way. The radio can learn which packets form one frame and how much each frame matters [20], and an application can be told what rate the cell can carry [19]. Figure 1 draws the resulting control loop. Every mechanism in this review closes some part of it, and the question for each is whether the part it closes is the one users perceive.

This review covers work published from 2023 to 2026 and follows the loop from the user outwards: what users perceive (Section 2), what measurements of real systems show (Section 3),

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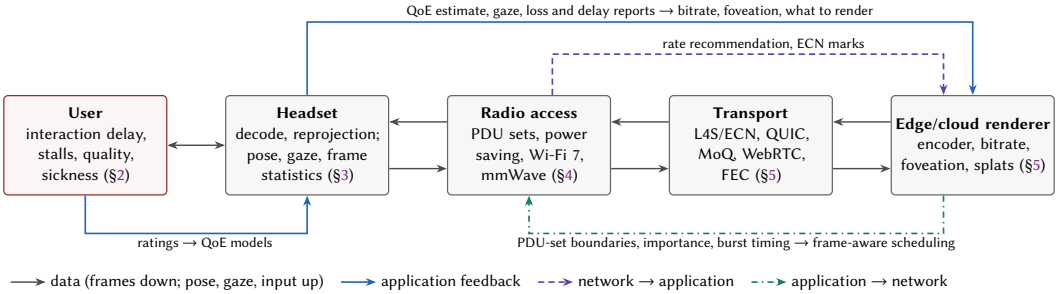


Fig. 1. The QoE control loop of a networked XR session and the sections that discuss each part. The application adapts to its own measurements, the network recommends rates or marks congestion, and the application tells the network which packets form a frame. Only the path from users' ratings to QoE models rests on user studies.

the radio access (Section 4), and transport and delivery (Section 5). Table 1 classifies the families of mechanisms, and Section 6 states open problems. Throughout, we separate what is *established* (measured on real systems or with users), what is *contested*, and what is *assumed*.

2 WHAT USERS EXPERIENCE

2.1 Loss and jitter outweigh delay

Recent user studies agree on one ordering: loss, jitter and stalls hurt more than tens of milliseconds of added delay. With Nvidia CloudXR and a first-person shooter (N=30), QoE fell at a round-trip time (RTT) of 75 ms or more, or at loss above 6%; combined, 25 ms with 4% loss was already harmful [47]. In a social XR application (N=28), RTT up to 77 ms had no significant effect, while loss above 2% at higher RTT did, and even jitter of one standard deviation degraded QoE at RTTs beyond 52 ms [37]. With a Quest 3 streaming over Air Link (N=16), loss harmed experience more than delay across almost all measured aspects [53]. In Rec Room and VRChat, the virtual environment froze under loss but not under latency [49].

The ordering follows from how the headset splits latency. Reprojection serves head rotation locally, so network delay reaches only *interaction latency*: aiming, grabbing, or seeing a peer react. That budget depends on timing structure, not on RTT alone. In split-rendered gaming, 90 and 120 fps tolerated up to 6 ms of network delay before responsiveness suffered, and 60 fps was the better choice when delay was high [35]. The budget is therefore *contested*: requirement statements still ask for sub-20 ms end-to-end latency [15], while users of cloud and social XR tolerated RTTs near 75 ms. Cybersickness depends on content more than on the network: loss raised sickness in Beat Saber but lowered presence in Cubism [53], and at most 21% of viewers reported dizziness for point clouds in AR [39].

2.2 Experience is social

Multi-user sessions add effects no single-user study can show. In collaborative VR (20 participants in pairs), users told impaired from unimpaired sessions but not latency from bursts, task time and sickness scores did not change, and the user's role shaped perception [12]. Impairments also raised synchronisation traffic four- to fivefold through retransmissions. Avatars expose a fidelity trade-off. With 40 participants on Quest Pro headsets, a light cartoon avatar gave the highest QoE under impairment; a humanoid avatar that replicates facial expressions lost more under delay, yet seeing

a partner's face softened the loss [5]. Standards lag behind: ITU-T leads in modelling social aspects of QoE, but 3GPP, ITU-T and ETSI efforts remain fragmented and centred on system metrics [38].

2.3 Predicting experience

Three families of models map what can be observed to what users would report. *Opinion-trained models* fit ratings from a lab study: cloud VR gaming models reached a Pearson correlation of 0.93 [30], and regression models for a VR shooter a mean absolute error of 0.14 [46]. Fed with real traces, the latter predicted a mean opinion score (MOS) of only 2.5–3.0 over 4G and 5G against 4.2–4.4 over wired access. *Content-quality metrics* cover new media: colour PSNR correlated with point-cloud ratings at 0.84 [39], but common metrics missed the effect of viewing behaviour on volumetric video [55]. *Network-side estimators* infer QoE from what an operator sees, such as encrypted 360-degree traffic over HTTPS and QUIC in 4G and 5G drive tests [24] or QoS counters on an O-RAN testbed [28], but are validated against application metrics, not users.

All three families inherit one study design: *static impairments*. Emulators apply a constant RTT, independent loss or a fixed jitter distribution to 16–40 participants, and real 4G and 5G measurements enter only afterwards, as inputs to models fitted under constant conditions [37, 46]. That thresholds learned this way hold under bursty, correlated wireless loss is *assumed*.

3 MEASURING REAL SYSTEMS

3.1 Commercial headsets and platforms

Measurements of commercial systems show that the application stack, not the link, often decides what users see. On Apple Vision Pro, FaceTime sends a spatial persona over QUIC at about 0.7 Mbps; the bitrate matches face and hand keypoints, not a mesh, which would need over 100 Mbps at 90 fps [10]. The same study found no rate adaptation: with the uplink capped at 0.7 Mbps the persona disappeared. Servers were chosen near the session initiator, which gave RTTs above 100 ms within the United States, and downlink traffic grew almost linearly from two to five users because servers forward each persona. On a Quest Pro, the old 2.4 GHz Wi-Fi 4 kept 99% of samples at or above 65 Hz against a 72 Hz target, while wireless offloading drained the battery 15% faster than local rendering [25]. One 45-minute social VR session on Multiverse touched 87 service domains, and its time-critical state ran as UDP at about 30 packets per second upstream [33]. Steam Link on a Quest 3 carries video and control in one QUIC tunnel, with a bursty downlink and a periodic uplink; scene complexity set the mean throughput, the player's style its variance [40].

3.2 Access networks in practice

In-situ tests expose interactions that link-level figures hide. One Wi-Fi 6 access point served three headsets at up to 100 Mbps each with latency near 50 ms, but uplink OFDMA delayed tracking packets and thus rendering [36]. With Unity Render Streaming over Wi-Fi 6, the WebRTC pacer released frames in 5.56 ms batches and added 6–21 ms of frame assembly delay although the RTT was 1.5 ms [7]. Cellular access trades direction: a 5G standalone edge setup gave a steadier downlink and better mobility than Wi-Fi 6, but higher uplink latency on the control channel [45], and a local 5G test network measured 3.4–4.1 ms one-way delay downlink against 6.4–14.8 ms uplink [15]. In these studies the uplink, which carries pose and input, was the weaker direction.

3.3 Testing the traffic models

The simulations in Section 4 rest on traffic models, and measurements now test them. For cloud VR on a Pico Neo 2, the 3GPP baseline underestimated the effective bitrate by 25% [27]. Frame sizes in ALVR depend on head angular velocity, so a model that includes motion predicted them

Table 1. Families of network and delivery mechanisms for XR since 2023, what they were evaluated on, and whether users confirmed the gain. "Proxy" means a network or image metric stands in for QoE.

Family	Idea	Evaluated on	QoE validated?
Frame-aware scheduling	Schedule PDU sets against a frame deadline	System-level simulation	No; PDB met
Coexistence and power	Limit interference; sleep between frames	System-level simulation	No
Network-assisted rate	Network recommends a rate	Simulation, testbed	No; outage
Wi-Fi 7 MLO, mmWave	Several links; beams that follow the head	Simulation, lab hardware	No; delay, loss
Early congestion signals	L4S/ECN, in-network dropping	Emulation, testbeds	Proxy; frame loss
QUIC and MoQ delivery	Streams, delivery timeouts	Emulation	Proxy; stalls
Selective protection	FEC on what matters, neural repair	Loss traces	Proxy; freezes
Perception-driven sending	Foveation, QoE-model encoding	Prototypes with users	Yes, small labs
Volumetric, splat delivery	Layered, progressive, shared content	Datasets, prototypes	Mostly proxy

better [56]. Modern 6DoF gaming reached 251.9 Mbps and 120 Hz, beyond the fixed 50 Mbps and 60 Hz many models assume [11], and cloud-rendered AR traces showed heavy-tailed packet sizes and loss that is rare on average but arrives in bursts [29]. Real XR traffic is larger, burstier and more motion-dependent than baseline models assume, so the capacity figures of Section 4 are likely optimistic.

4 THE RADIO ACCESS

The common idea of radio work since 2023 is to change the unit of service from the packet to the video frame. A frame that misses its deadline is worthless even if most of its packets arrive. Table 1 classifies the resulting families together with those of Section 5.

4.1 Making the radio frame-aware

PDU sets are the standard vehicle. 3GPP Release 18 defines a PDU set as the smallest group of packets that carries one frame, together with its relative importance, and lets the radio network apply QoS handling to it [20]. Release 19 adds burst signalling, expedited transfer of latency-critical content and guidance on importance values [21], plus radio features such as latency-aware uplink prioritisation, delay status reports and uplink rate control through application adaptation [41]. *Frame-aware schedulers* exploit this information. A PDU-set scheduler raised XR capacity by up to 45% at unchanged broadband throughput [44], and an uplink least-slack-time scheduler gained 15% capacity and cut XR packet delay by up to 33% in a nine-cell dense urban case [13].

4.2 Coexistence and energy

XR rarely has a cell to itself. Adding a 45 Mbps XR flow with a 10 ms packet delay budget (PDB) cost about 45 Mbps of broadband capacity [43], but interference from broadband users in other cells removed about 80% of XR capacity; restricting broadband throughput by about half restored it at a 20 ms PDB [42]. Power saving competes with delay: standardised and proposed sleep schemes saved 6–23% of device energy without delaying traffic unacceptably [51].

4.3 Telling the application what the network can carry

A second family closes the loop from network to application (Figure 1). NetXRate combines core-network analytics with an O-RAN xApp to recommend rates, and cut outage by up to 96% against over-the-top adaptation in simulation [19]. On a 4G O-RAN testbed, an xApp held VR latency near its target and freed 15.8% of resources for other users [8].

4.4 Wi-Fi 7 and millimetre wave

Multi-link operation (MLO) in Wi-Fi 7 lets one device use several channels. In simulation, it supported seven AR stations where a single link supported six, while policies that need simultaneous access to all links suffered under contention [3]; for the same total bandwidth, more and narrower links gave lower delay than fewer wide ones [6]. *Millimetre wave* offers the rates that uncompressed Quest 3 video would need (15.75–26.25 Gbps), yet tested hardware delivered 1.85–1.98 Gbps, and moderate head rotation at 45 degrees per second drove loss to 4.72% in one configuration [48]. Dual-beam reception across coordinated access points improved reliability by up to 22.8% on real head-motion traces [23].

These results share their assumptions: system-level simulation with 3GPP-style traffic, which Section 3 found too light, and a user counted as satisfied when enough packets meet the PDB. No study has yet shown that frame-aware scheduling raises the QoE that Section 2 measured.

5 TRANSPORT AND DELIVERY

Above the radio the question is the same: deliver each frame before its deadline, and send less when the path cannot.

5.1 Earlier congestion signals

Delay- and loss-based controllers react after a queue has formed, which is a frame too late for XR. *L4S* marks congestion before queues grow. In volumetric video conferencing, an L4S pipeline converged on the available rate 45% faster than Google Congestion Control (GCC) and lost no packets, while the WebRTC baseline left 3% of frames undecodable [16]. *Comparisons of existing controllers* remain mixed: integrated into ALVR, GCC tracked the available rate closely with few dropped frames, but only NADA held a fair share against loss-based flows [2]. *In-network adaptation* moves the decision to the base station. Octopus lets senders overestimate capacity and drops application-chosen packets at the cell, reporting 1.5–18× better performance for standard and volumetric video [9].

5.2 Transports with deadlines

QUIC removes transport head-of-line blocking and offers unreliable datagrams. *Media over QUIC* (MoQ) adds delivery timeouts, which a point-cloud server used for implicit adaptation: clients with tight latency targets received lower quality, relaxed ones higher quality [18]. WebRTC keeps its strengths for groups: through a selective forwarding unit, volumetric video adapted to user movement 13% and 36% faster than over TCP at 5 and 10 ms of network latency [17]. Which transport suits interactive XR is still *contested*.

Section 2 showed that loss, not delay, drives QoE, yet a retransmission costs a frame time. *Selective protection* spends redundancy where it matters. ReVo applies forward error correction (FEC) to critical colour and depth content and repairs the rest with a neural module, cutting freezes by up to 95.7% on real loss traces [1]. A hybrid broadcast–unicast design masked typical wireless loss with 15% FEC overhead while keeping volumetric latency under 40 ms [22].

5.3 Adapting what is rendered and sent

Application-layer rate adaptation is the oldest loop and still effective: over Wi-Fi, a network-aware bitrate algorithm for ALVR adapted to contention and mobility [34]. *Perception-driven sending* lets the user decide what to send, and it is the one family validated with users. Dynamic foveation driven by streamed gaze added 0.60 MOS over static foveation while saving 9.81% bitrate in cloud VR gaming [14], and EyeNexus adapted the foveal region to bandwidth and cut latency by up to 70.9% [54]. Encoding settings driven by a learned QoE model raised interaction quality by 1.20 MOS

[30]. *Split decisions* move work between device and edge: RenderFusion switches objects between local and remote rendering using a perceptual study of latency [32].

5.4 Volumetric and Gaussian-splat delivery

Volumetric content turns the trade-off between bandwidth and latency into one between compression and latency. In VR2Gather, compression cut bandwidth by more than 90% but raised latency to 278.7 ms, and DASH delivery reached 562 ms against 87.4 ms over TCP [50]. MuV2 served over 30 untethered devices from one access point by mixing direct volumetric streaming with remote rendering [31]. Gaussian splats bring compact, progressive formats: V^3 streams about 0.5 MB per frame and renders at 27 fps on a phone [52], Vega reaches 30 fps on smartphones by prioritising important objects [26], and TIGAS renders splats remotely and streams 2D views over QUIC [4]. These systems report PSNR, SSIM and frame rate; none of them reports a networked user study.

6 OPEN PROBLEMS AND CONCLUSION

The loop in Figure 1 is now closed at every layer, but each layer closes it against its own metric: the radio against delay budgets, transport against frame loss, the renderer against image quality. Five questions would join the parts.

Open problems

- **P1. Do QoE thresholds survive real network dynamics?** Every user study in Section 2 applied static delay, independent loss or fixed jitter, while measured wireless loss arrives in bursts [29]. Replaying recorded Wi-Fi and 5G traces to participants would show whether the tolerance of 75–77 ms RTT and the dominance of loss still hold.
- **P2. Does frame-aware scheduling raise QoE?** Frame-aware schedulers report 15–45% more capacity [13, 44], counted as packets that meet a PDB under traffic models that underestimate real demand [27]. A testbed that couples such a scheduler to a headset and a QoE model would test whether capacity gains are experience gains.
- **P3. Who should own the rate?** Network recommendations [19], Release-19 uplink rate control [41], L4S marks [16] and application bitrate adaptation [34] can all act on one flow. Their joint stability, fairness and response time are unstudied.
- **P4. What is group QoE?** Social studies test pairs [5, 12], while platforms serve five or more users through forwarding servers [10]. How delay assigned to one member changes the experience of the others, and how to share impairments fairly, is unknown.
- **P5. Which representation for which link?** Point clouds, keypoint avatars, Gaussian splats and remotely rendered video trade bandwidth, device computation and latency differently, but each is evaluated alone and on image metrics [4, 26, 52]. A common user-facing benchmark over the same network traces is missing.

Since 2023, networked XR has gained frame-aware radios, earlier congestion signals, transports with deadlines and content built for adaptation, while users have shown that reliability matters more than the last tens of milliseconds. The next step is to judge each mechanism by the experience it produces under real network dynamics, alone and in groups.

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